Hemophagocytic Lymphohistiocytosis

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Definition

- ☑ Syndrome characterized by severe systemic hyperinflammation which clinically presents with unremitting fever, cytopenias, organomegaly (hepatosplenomegaly), and raised inflammatory markers. Patients can develop liver failure, neurological issues, coagulopathy, and multiorgan failure.
- ☑ If untreated or there is a delay in diagnosis, the syndrome can have a high mortality rate.

Causes

Hemophagocytic lymphohistiocytosis (HLH) can be caused by either mutations that are inherent to the individual which cause increased macrophage activation (primary HLH) **(Table 1)** or due to underlying secondary causes which can range from infections, rheumatological disorders, malignancies (secondary HLH**) (Table 2)**.

Causes

TABLE 1: Genetic hemophagocytic lymphohistiocytosis (HLH) (primary HLH).

Familial HLH (FHLH)	***Gene***	***Chromosome***
FHLH1	Unknown	9q22.1-23
FHLH2	PRF1	10q22
FHLH3	Unc13	17q25
FHLH4	STX11	6q24
FHLH5	STXBP2	19p13
Pigmentary disorders associated with HLH		
Griscelli syndrome type 2	RAB27A	15q21
Chédiak–Higashi syndrome	LYST	1q42.1-42.2
Hermansky–Pudlak type II	AP3B1	5q14.1
Syndromes associated with HLH		
XLP1	SAP	Xq25
Lysinuric protein intolerance	SLC7A7	14q11.2
XLP2	XIAP	Xq25

TABLE 2: Causes of secondary HLH.

Infections	***Malignancies***	***Rheumatological disorders***	***Metabolic disorders and inborn errors of immunity (IEI)***
☑ *Viruses:* EBV, CMV, HHV6, HHV7, HSV, hepatotropic viruses, dengue, parvovirus B19 ☑ *Bacteria: Salmonella, Brucella, Leptospira, Mycobacterium* spp. ☑ *Protozoa: Leishmania* ☑ *Fungi: Histoplasma, Cryptococcus*	☑ *Leukemias:* T > B cell ☑ *Lymphomas:* – T and NK cell lymphoma – Anaplastic large cell lymphoma – Diffuse large B cell lymphoma – Hodgkin lymphoma	☑ Juvenile idiopathic arthritis ☑ PAN ☑ Systemic lupus erythematosus ☑ Kawasaki disease ☑ NLRC4 inflamma-somopathies	☑ *Metabolic disorders:* – Lysinuric protein intolerance – Biotinidase deficiency – Wolman disease – Gaucher disease – Pearson syndrome ☑ *IEI:* – SCID, DiGeorge syndrome, Wiskott–Aldrich syndrome, ataxia–telangiectasia, dyskeratosis congenita

(CMV: cytomegalovirus; EBV: Epstein–Barr virus; HLH: hemophagocytic lymphohistiocytosis; NK: natural killer; PAN: polyarteritis nodosa; SCID: severe combined immunodeficiency)

HLHS vs MAS – Two Sides of the Same Coin?

Rheumatologists encounter HLH in patients with systemic juvenile idiopathic arthritis. HLH in this setting is called as macrophage activation syndrome (MAS). The criteria encompass similar variants as that of the histiocyte society but different cutoffs are used. Both disorders present with hyperferritinemia, high fever, cytopenias, and organomegaly, both if untreated can lead to severe coagulopathy, multiorgan dysfunction, and finally death. The mainstay of therapy for MAS is usually treatment of the underlying disorder with potent immunosuppressive agents whereas HLH is usually treated with a combination of immunosuppression, chemotherapy and if severe or familial, hematopoietic stem cell transplantation.

Box 1 shows the European League against Rheumatism Guidelines for MAS.

BOX 1: The European League against Rheumatism Guidelines for macrophage activation syndrome (MAS).

Ferritin >684 ng/mL and two of the following:

- ☑ Platelet count ≤181 × 10^9/L
- ☑ Aspartate aminotransferase >48 U/L
- ☑ Triglycerides >156 mg/dL
- ☑ Fibrinogen ≤360 mg/dL

Diagnosis

Diagnosis of HLH is Made by the HLH 2004 Criteria

This criterion not only employs the genetic susceptibility to develop HLH but along with it clinical signs and symptomatology of the disease (fever and organomegaly), laboratory evaluation (cytopenia, hypertriglyceridemia or hypofibrinogenemia, hyperferritinemia, and increased soluble CD25) and immunopathology [evidence of HLH and increased natural killer (NK) cell function). A patient must satisfy five of the eight criteria to satisfy the diagnosis.

Diagnosis

Hemophagocytic Lymphohistiocytosis 2004 Criteria

- ☑ Genetic defect consistent with HLH or
- ☑ Five out of eight clinical and laboratory criteria fulfilled
  1. Fever
  2. Splenomegaly
  3. Cytopenia in ≥2 cell lineages
     - Hemoglobin <9 g/dL, in neonates <10 g/dL
     - Platelet count $<100 \times 10^9$/mL
     - Neutrophil count $<1 \times 10^9$/mL
  4. Hypertriglyceridemia (>265 mg/dL) or hypofibrinogenemia (<150 mg/dL)
  5. Hyperferritinemia (>500 ng/mL)
  6. Soluble CD25 >2,400 U/mL (or elevated compared with laboratory defined normal ranges)
  7. Hemophagocytosis in bone marrow, spleen, lymph nodes, or liver
  8. Low or absent NK cell cytotoxicity

Ferritin in Hemophagocytic Lymphohistiocytosis

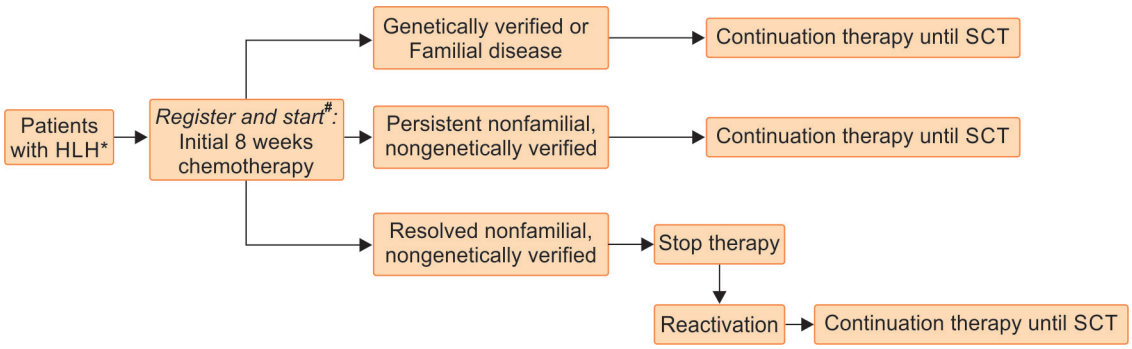
Important to note that all children should not be labeled as HLH based on just the serum ferritin showing a value of >500 ng/mL. Ferritin is an acute phase reactant and can be elevated during febrile phases. It is important to see the trend of rise of ferritin and correlate it with the other signs and symptoms of HLH. A ferritin value of >10,000 ng/mL is highly sensitive and specific for HLH.

Who should be Screened for Primary HLH Mutations?

- ☑ Patients with severe/persistent disease
- ☑ Patients with reactivation of disease
- ☑ Known family history of HLH/recurrent pregnancy losses
- ☑ Presentation in infancy

- ☑ Hemophagocytic lymphohistiocytosis treatment incorporates tempering down the hyperinflammatory state with the help of immunosuppressive agents (steroids) and combination of chemotherapeutic agents.
- ☑ Children with HLH are sick at initial presentation and many require intensive care support during that period. Organ-specific effects of HLH need to be monitored and supportive care with inotropic support and extensive blood product transfusions may be required. Organ failures may need replacement therapy.
- ☑ In cases with secondary HLH, it is pertinent to remember that treating the triggering cause should go together with the treatment of the HLH. Treatment of HLH should always be in a unit that has pediatric intensive care support. Patients with severe disease or familial HLH need to be referred under the care of a specialized pediatric hematology oncology unit.
- ☑ Patient will need *Pneumocystis jirovecii* prophylaxis with trimethoprim-sulfamethoxazole and antifungal prophylaxis.
- ☑ The Histiocytic Society has laid down guidelines for treatment of HLH with its two successive trials—HLH 1994 and HLH 2004 studies **(Flowchart 1)**. Patients with central nervous system (CNS) disease need further intrathecal treatment with steroids and methotrexate **(Fig. 1)**.

Flowchart 1: HLH 2004 protocol—initial therapy—week 1–8.



(HLH: hemophagocytic lymphohistiocytosis; SCT: stem cell transplantation)

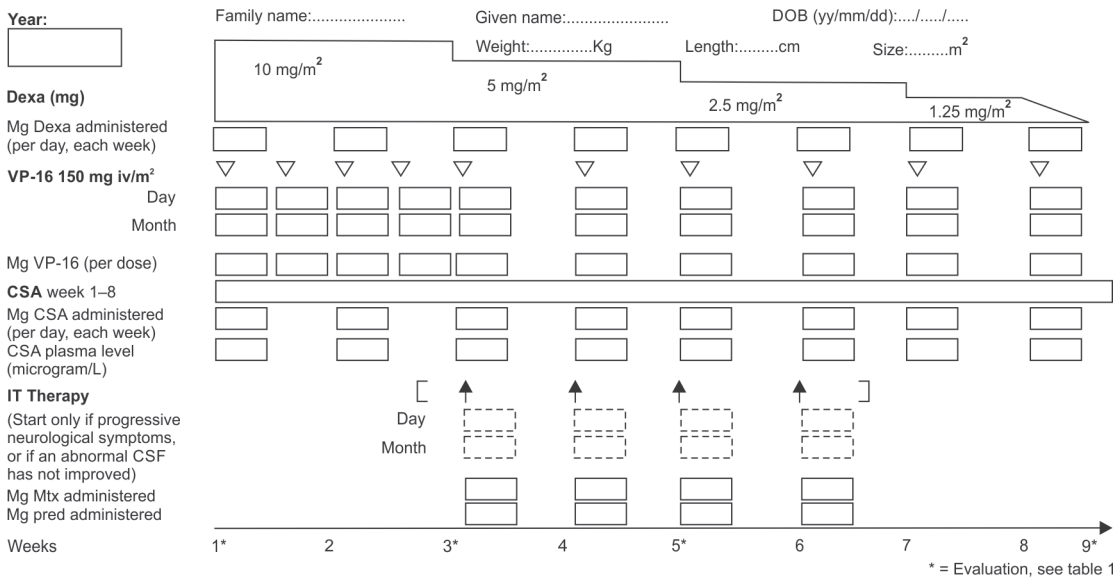


Fig. 1: Initial course—week 1–8.
(CSA: cyclosporin A; CSF: cerebrospinal fluid; IT: intrathecal)

- ☑ *Dexamethasone*: 0 mg/m² week 1–2, 5 mg/m² week 3–4, 2.5 mg/m² week 5–6, 1.25 mg/m² week 7, steroids are tapered and discontinued in week 8
- ☑ *Injection etoposide:* 150 mg/m² twice weekly for week 1–2, 150 mg/m² once weekly till week 8

  Contraindication: Absolute neutrophil count < 500/mm³ or hypocellular marrow
- ☑ *Cyclosporin A*: 6 mg/kg/day with monitoring of levels, target trough levels 200 µg/L
- ☑ *Intrathecal therapy:* Cerebrospinal fluid (CSF) evaluated at diagnosis and after 2 weeks. Persistence of progressive neurological symptoms or if an abnormal CSF (cell count and protein) has not improved, intrathecal therapy with 4 weekly injections is done.

  Dose:
  - *Methotrexate:* <1 year—6 mg, 1–2 years—8 mg, 2–3 years—10 mg, and >3 years—12 mg
  - *Prednisolone:* <1 year—4 mg, 1–2 years—6 mg, 2–3 years—8 mg, and >3 years—10 mg

Treatment

Continuation Therapy—From Week 9–40

- ☑ *Etoposide (VP 16):* 150 mg/m^2 iv every 14 days
- ☑ Dexamethasone pulses every 14 days, 10 mg/m^2 for 3 days
- ☑ *Cyclosporin A:* Aim for blood levels around 200 µg/L

Patients with primary HLH/reactivation subsequently move to hematopoietic stem cell transplantation (HSCT).

Transplant in Hemophagocytic Lymphohistiocytosis

- ☑ Matched sibling donor, matched unrelated donor, haploidentical donor, and cord blood transplants have all been described for HLH.
- ☑ Myeloablative conditioning was previously used (busulfan, cyclophosphamide, and etoposide), and has high potential of treatment related toxicity [increased risk of veno-occlusive disease (VOD)] and inferior survival of around 40–50%.
- ☑ Reduced toxicity conditioning with treosulfan, fludarabine, thiotepa/alemtuzumab, treosulfan, and fludarabine leads to stable mixed chimerism and improved survival in tune of 70–80%.
- ☑ It is important to screen the sibling with genetic studies for nascent homozygous disease before going forward with the transplant (heterozygous carriers are accepted).

Further Reading

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Cleft Lip and Palate

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